



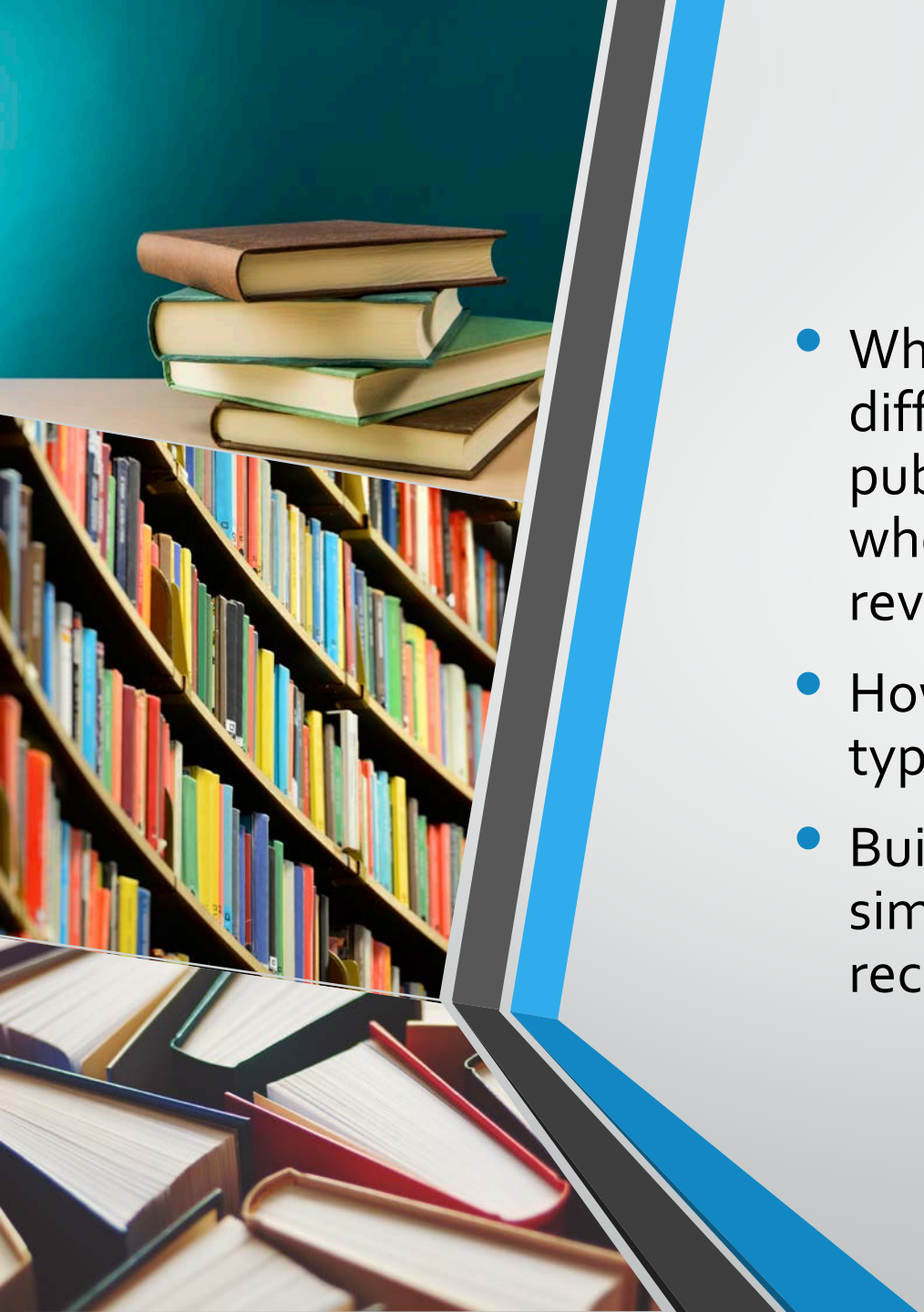
EBOOK RECOMMENDER SYSTEM

DATA SCIENCE CAPSTONE THREE PROJECT

MINGYU (BERT) LIU

PROBLEM

- When readers choose books, they are influenced by different factors such as the type of book, author, publisher, price, whether it is recommended by editors, whether it is a popular book, and other readers' reviews.
- How can we recommend suitable books to different types of readers?
- Build a recommendation model based on content similarity or clustering to provide personalized book recommendations to readers.



WHO MIGHT CARE?

E-commerce Platform



Bookstore



Publisher

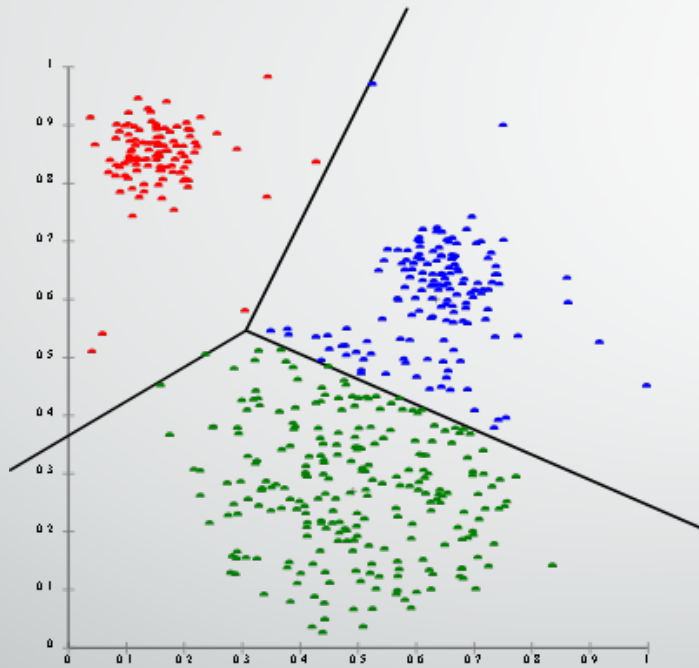


Readers

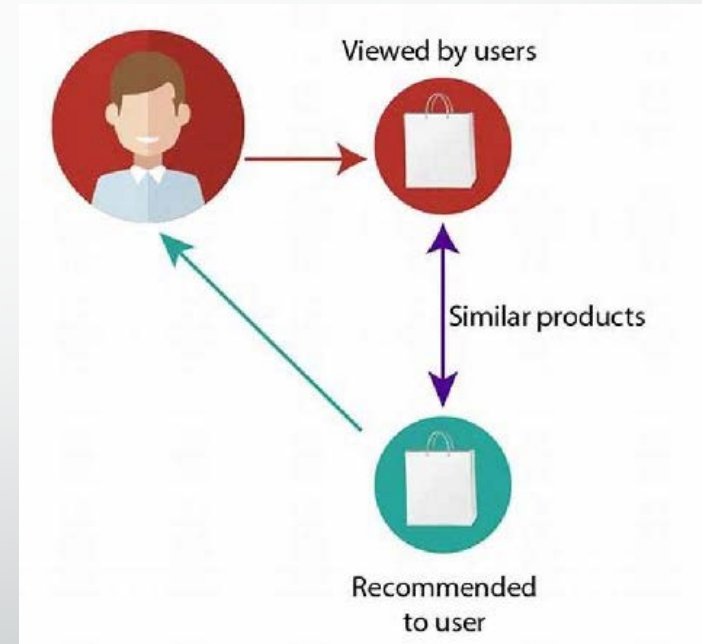


MAIN CONCLUSION

Based on the data, compared with the k-means cluster algorithm, the content-based filtering algorithm is more effective.

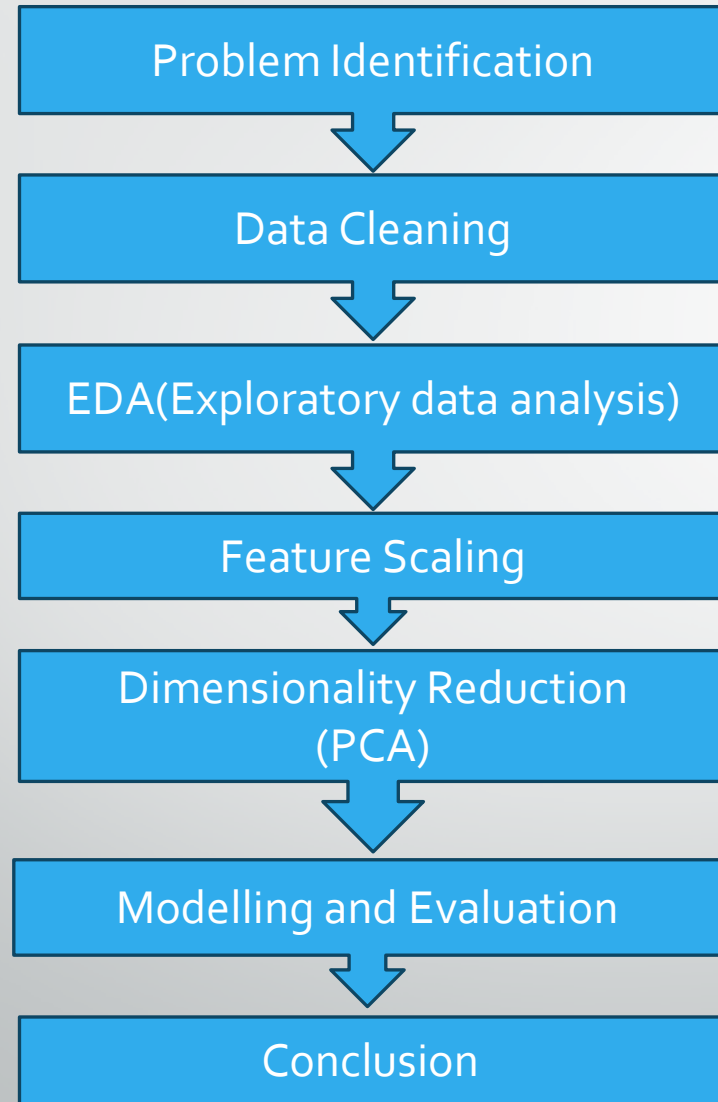


Clustering algorithms



Content-based filtering

THE DATA SCIENCE METHOD(DSM)



This is an unsupervised learning problem

Build Ebook Recommender System

K-means Algorithm
K-means Adjustment
Content-based filtering

DATA INFORMATION

Kaggle.com/ Amazon Kindle Books
Dataset

Amazon Kindle Books Dataset 2023
(130K Books)

RangeIndex: 133102 entries, 0 to 133101

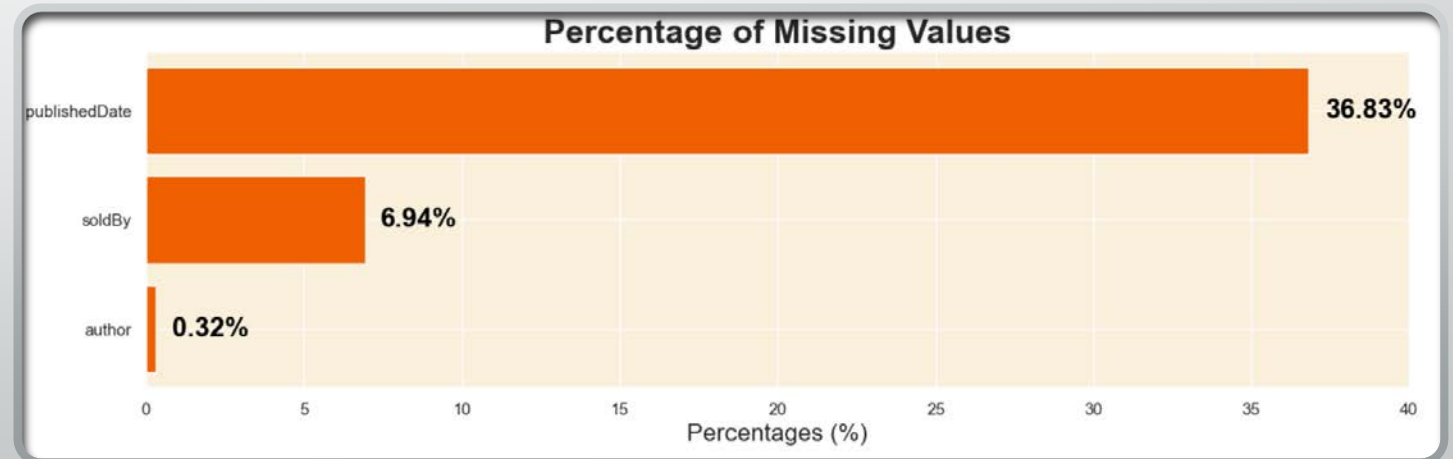
Data columns (total 16 columns):

#	Column	Non-Null Count	Dtype
0	asin	133102 non-null	object
1	title	133102 non-null	object
2	author	132677 non-null	object
3	soldBy	123869 non-null	object
4	imgUrl	133102 non-null	object
5	productURL	133102 non-null	object
6	stars	133102 non-null	float64
7	reviews	133102 non-null	int64
8	price	133102 non-null	float64
9	isKindleUnlimited	133102 non-null	bool
10	category_id	133102 non-null	int64
11	isBestSeller	133102 non-null	bool
12	isEditorsPick	133102 non-null	bool
13	isGoodReadsChoice	133102 non-null	bool
14	publishedDate	84086 non-null	object
15	category_name	133102 non-null	object

dtypes: bool(4), float64(2), int64(2), object(8)

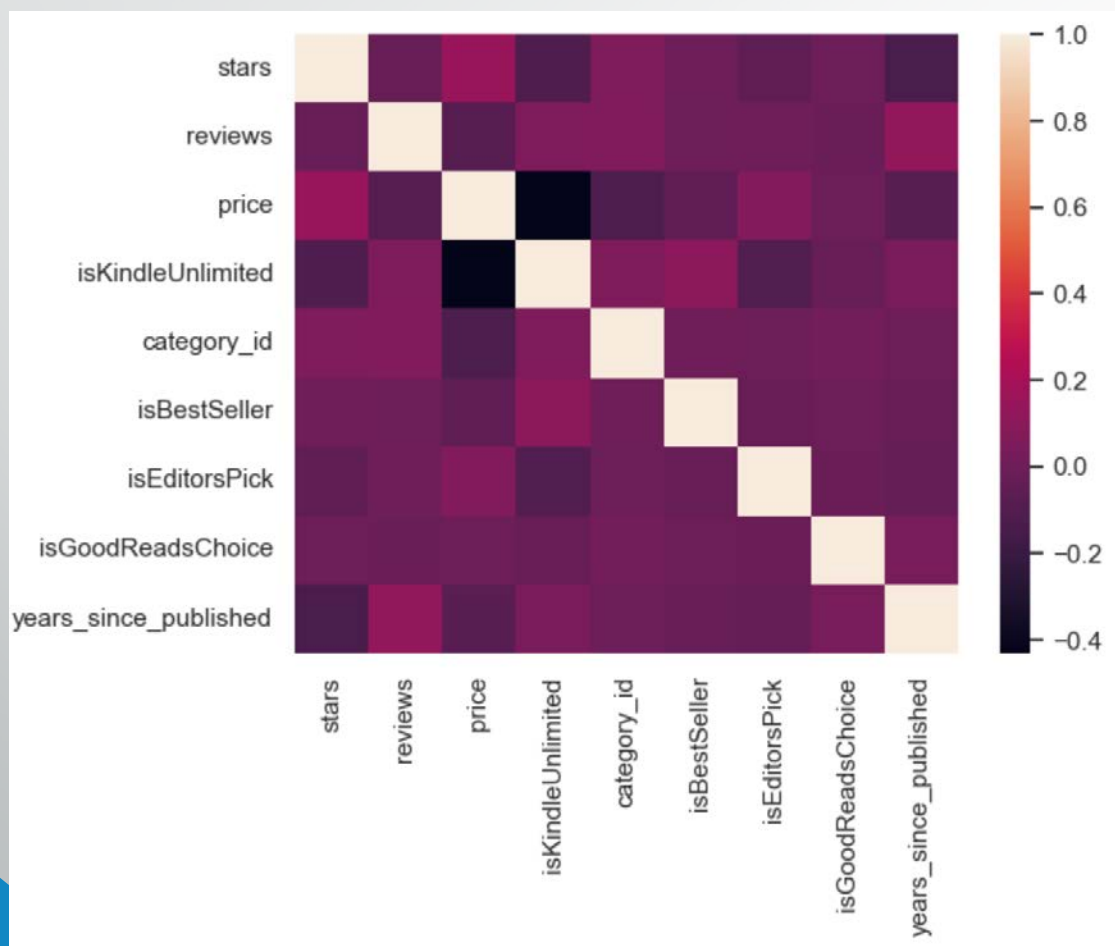
DATA CLEANING

- Handle Missing Values: Decide on strategies to fill missing values in numerical and categorical columns.
- Convert Data Types: Convert published date from an object to a datetime type.
- Outlier Detection: Identify and potentially remove or adjust outliers in numerical columns like price
- Checking and handling duplicates

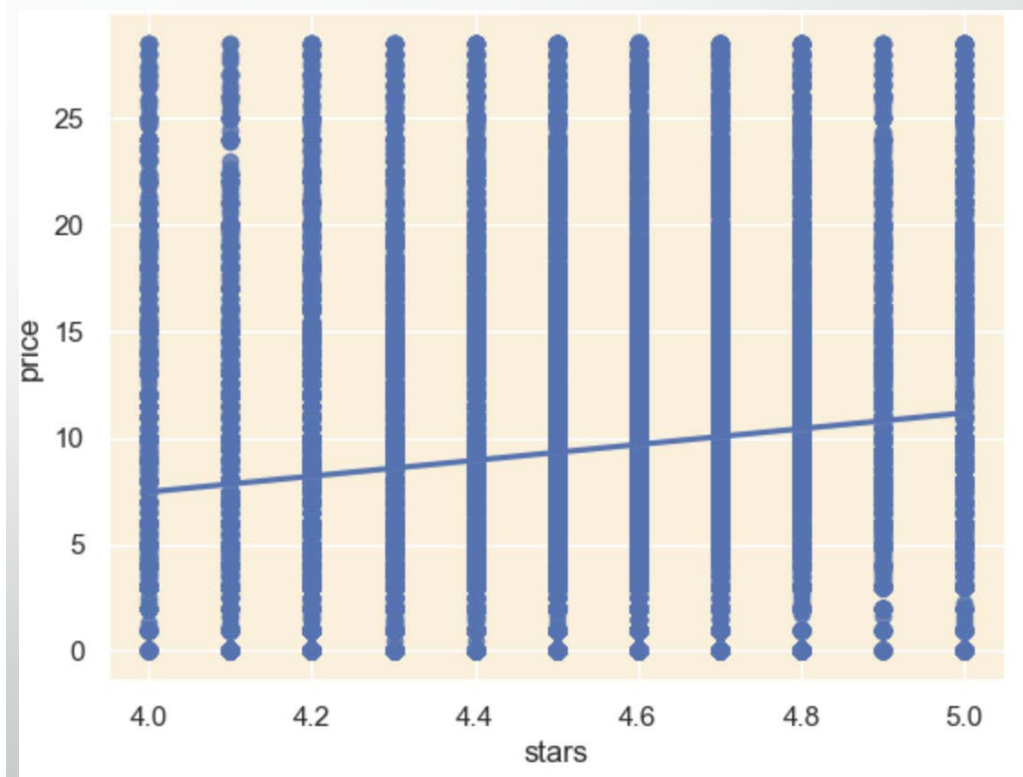


EXPLORATORY DATA ANALYSIS

Correlation matrix



Price and stars



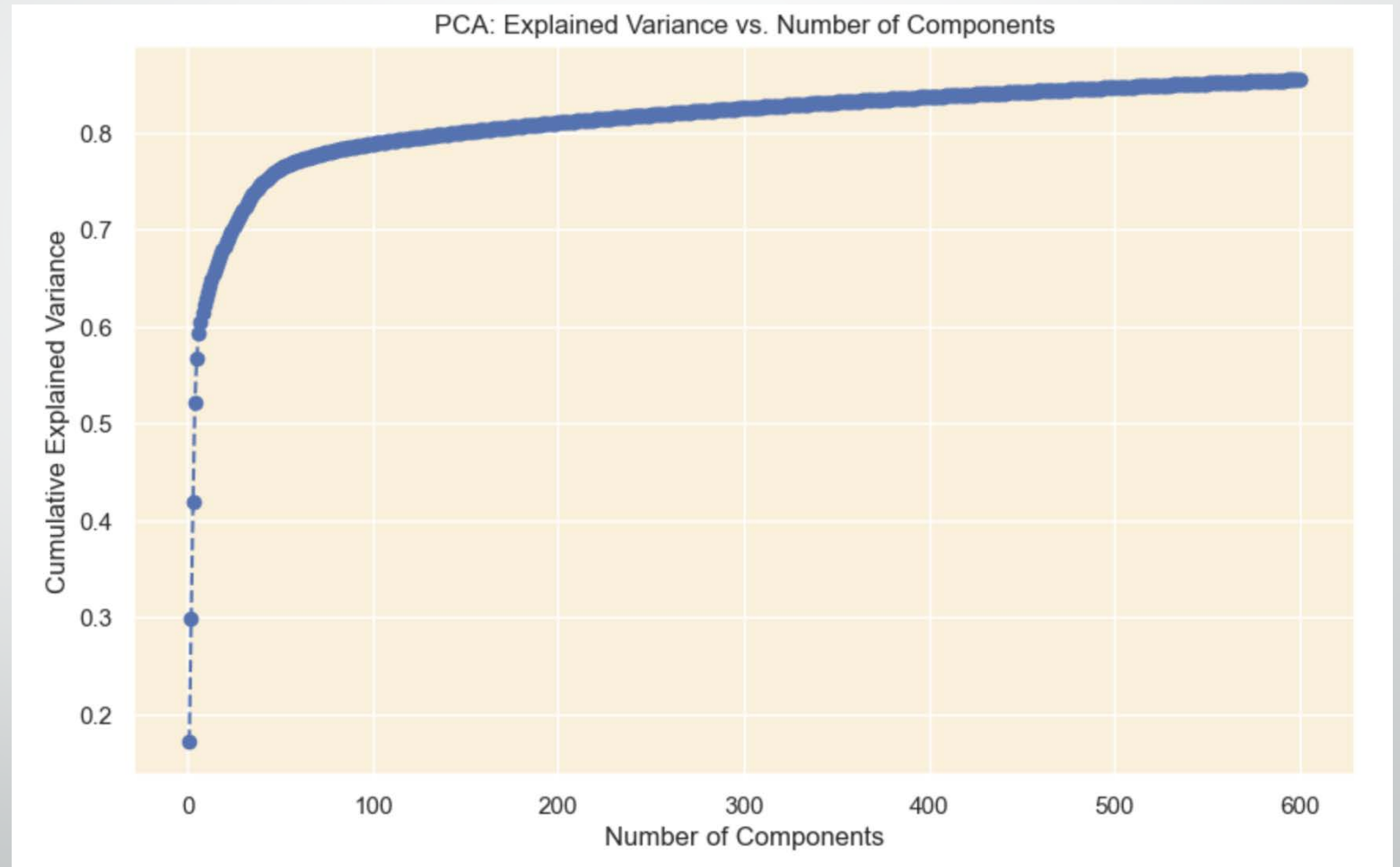
FEATURE SCALING

- from
sklearn.preprocessing
import StandardScaler

```
author_"Termite" Terry Singleton  author_'El Pela' Gonzalo Romero  \  
0                                0                                0  
1                                0                                0  
2                                0                                0  
3                                0                                0  
4                                0                                0  
  
author_(Chef) Dominique Ansel  author_(John D.) J. D. Turner  \  
0                                0                                0  
1                                0                                0  
2                                0                                0  
3                                0                                0  
4                                0                                0  
  
author_(Journalist) David Epstein  author_(Journalist) Leigh Gallagher  \  
0                                0                                0  
1                                0                                0  
2                                0                                0  
3                                0                                0  
4                                0                                0  
  
author_(Musician) KRS-One  author_(Psychotherapist) Richard O'Connor  \  
0                                0                                0  
1                                0                                0  
2                                0                                0  
3                                0                                0  
4                                0                                0  
  
author_(日)高野和明  author_, Annie's  ...  category_id_30  category_id_31  \  
0                    0                    0  ...  0                    0  
1                    0                    0  ...  0                    0  
2                    0                    0  ...  0                    0  
3                    0                    0  ...  0                    0  
4                    0                    0  ...  0                    0
```

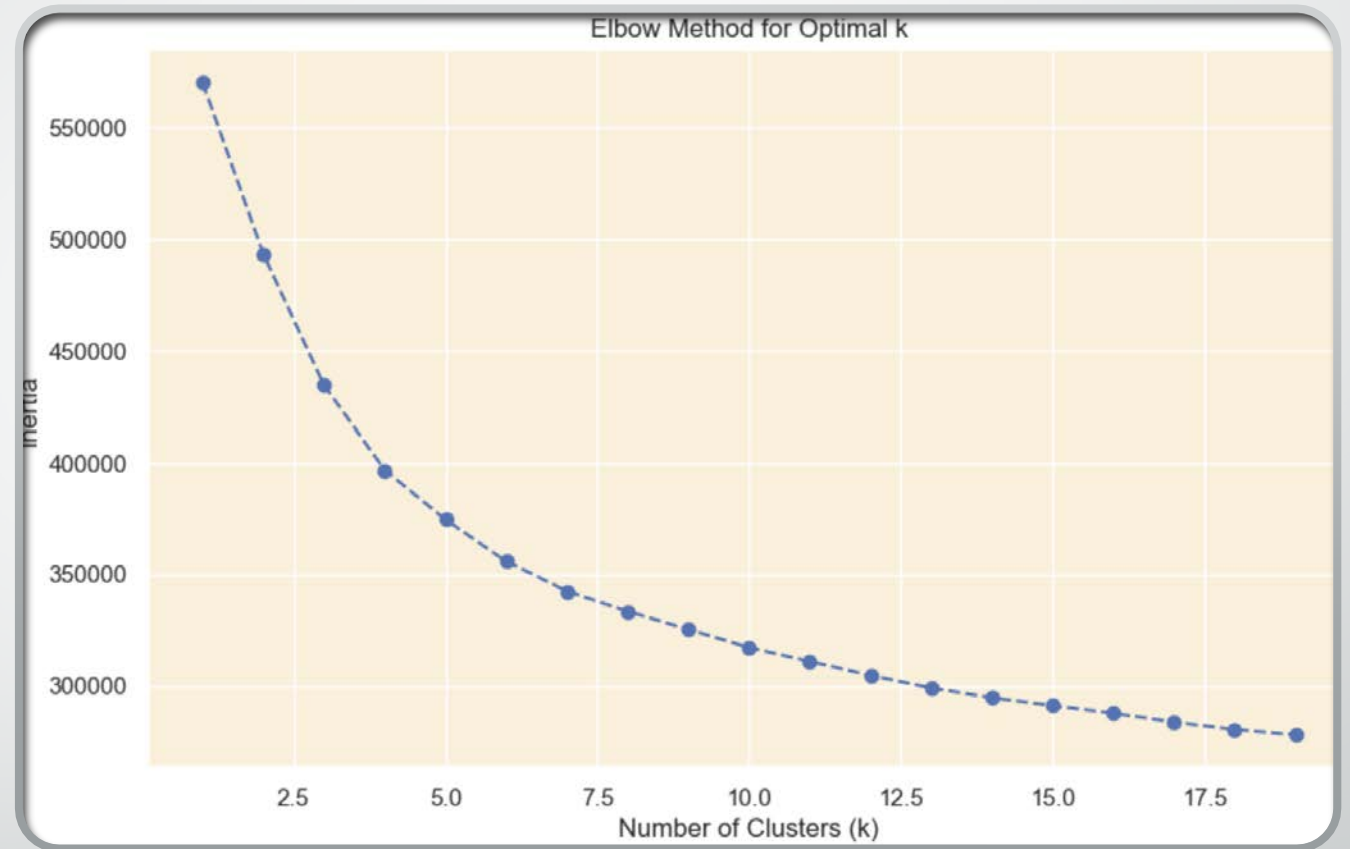
DIMENSIONALITY REDUCTION

- Initialize PCA with 600 components

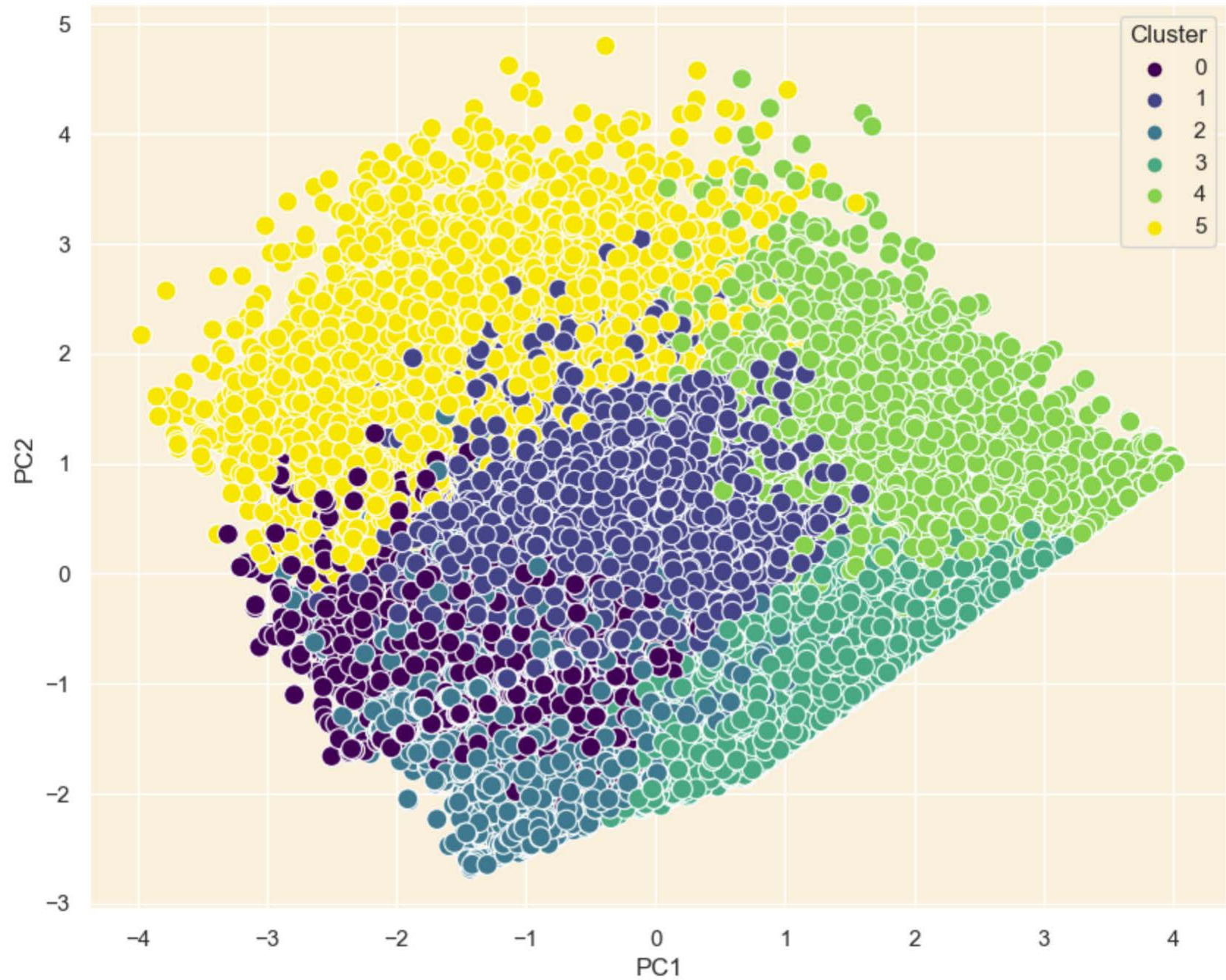


MODELLING AND EVALUATION

- k-means Clustering

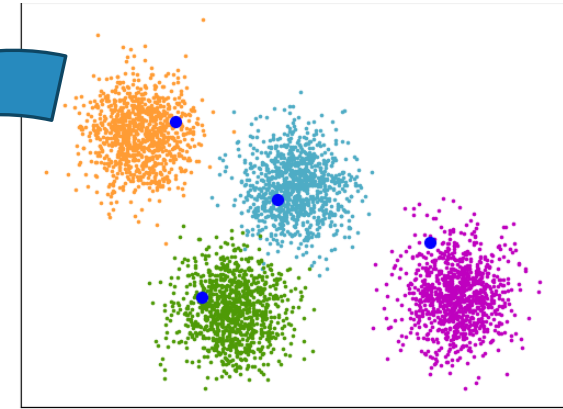


K-Means Clusters Based on First Two Principal Components



BUILDING RECOMMENDER SYSTEM-- CLUSTERING

- Based on clustering, recommend books in the same cluster(n_clusters=6).

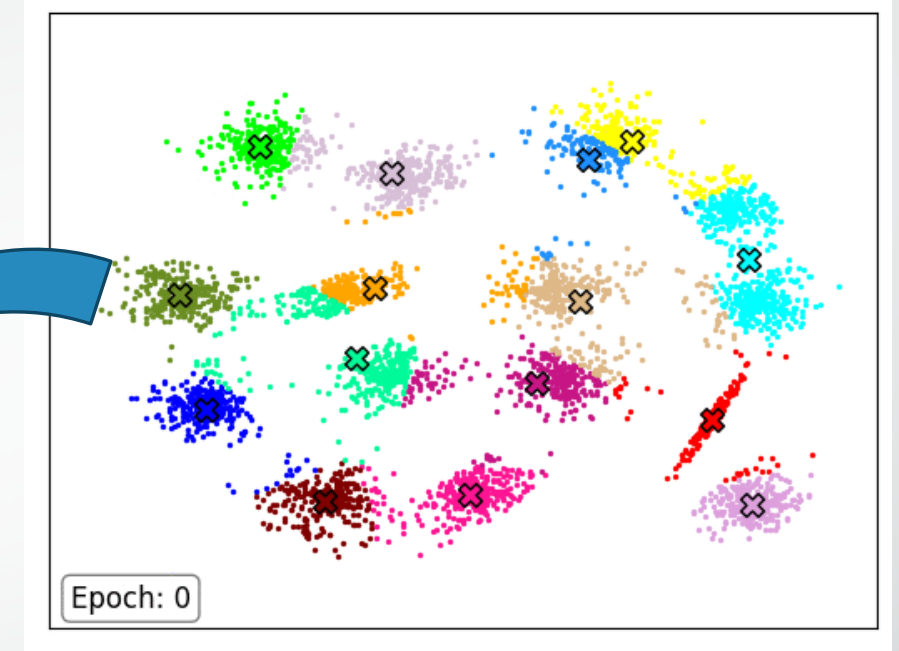


	title
74881	Crónicas Germanas: Tomo I (Casos reales docume...
1828	Black Women and Narcissists: Refuse to be Abus...
3582	Весь Александр Дюма (Russian Edition)
22350	Asentir: La magia de mi alma rota (Spanish Edi...
22181	Lazos de Sangre: Memorias de una Familia Salva...

	author	stars
74881	Igor Franz Wolff	5.0
1828	Nya Love	5.0
3582	Александр Дюма	5.0
22350	Brenda Amalia Soto Garcia	5.0
22181	Lisbeth Cornejo	5.0

BUILDING RECOMMENDER SYSTEM-- CLUSTERING

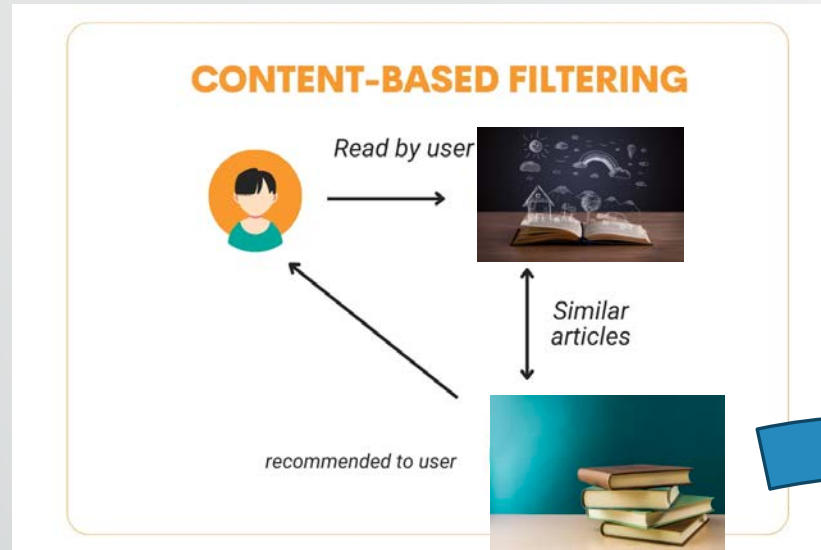
- Adjust parameters to divide into more clusters($n_clusters=20$).



	title	author
66405	Cook Korean!: A Comic Book with Recipes [A Co...	Robin Ha
65597	Danielle Walker's Against All Grain Celebratio...	Danielle Walker
13462	If This Book Exists, You're in the Wrong Unive...	Jason Pargin
65776	Amaro: The Spirited World of Bittersweet, Herb...	Brad Thomas Parsons
10259	The Longmire Defense: A Longmire Mystery	Craig Johnson

	stars
66405	4.8
65597	4.8
13462	4.8
65776	4.8
10259	4.8

BUILDING RECOMMENDER SYSTEM-- CONTENT-BASED FILTERING MODEL



- from `sklearn.metrics.pairwise`
`import cosine_similarity`

	title	author	stars
38	The State of Affairs: Rethinking Infidelity	Esther Perel	4.6
39	The Bullet Journal Method: Track the Past, Ord...	Ryder Carroll	4.6
59	Walking with Sam: A Father, a Son, and Five Hu...	Andrew McCarthy	4.6
79	The Book You Wish Your Parents Had Read: (And ...	Philippa Perry	4.6
98	Homecoming: Reclaiming and Healing Your Inner ...	John Bradshaw	4.6

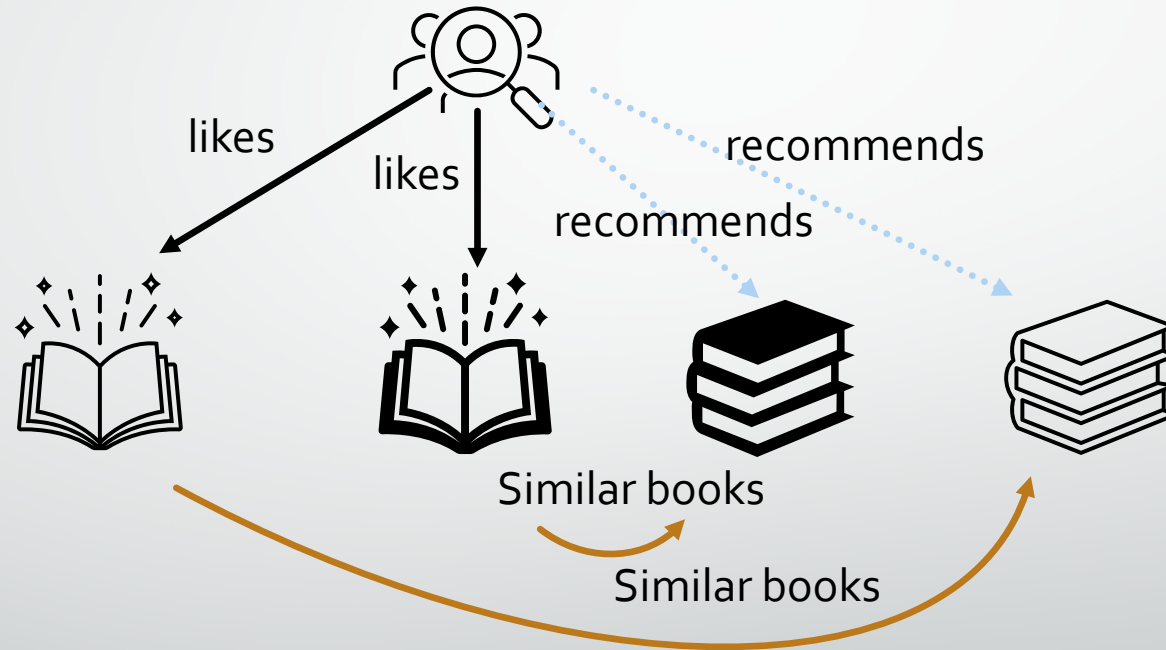
CONCLUSION AND RECOMMENDATIONS

- ❑ From the effect point of view, the content-based filtering algorithm is more effective.
 - Average Precision: 0.61
 - Average Recall: 0.46
 - Average F1-Score: 0.50
- ❑ The system can recommend several or more books to readers who have already purchased books of this type based on the similarity of the content, thus achieving more sales.

NEXT STEPS

- If more data on consumers' book purchases can be obtained, personalized recommendations can be made based on consumers' behavioral characteristics.
- If conditions permit, the recommendation system can be used in actual sales, and the effectiveness of the recommendation system can be further evaluated using A/B testing.

CONTENT-BASED FILTERING RECOMMENDER SYSTEM



Thank you for listening!